

COMPLEX COMPOUNDS

→ Double salt

- ① addⁿ of compounds, lose these properties in aq. medium.
- ② Partially hydrolysis,
- ③ Constituent ions are similar with properties. (aq)
- ④ due to metallic bond

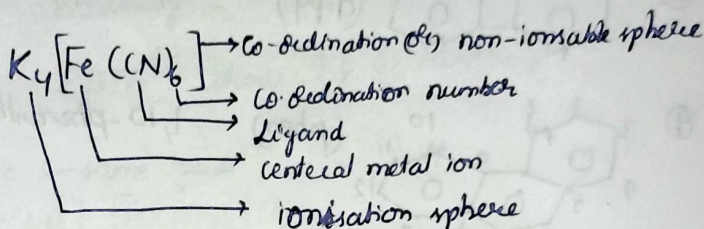
Ex: $KCl \cdot MgCl_2 \cdot 6H_2O$, $FeSO_4 \cdot (NH_4)_2SO_4 \cdot 6H_2O$

Complex salt

- ① addⁿ compounds, doesn't lose identity in aq. medium.
- ② Partial hydrolysis.
- ③ Constituent ions properties are different in (aq).
- ④ due to dative bond.

Ex: $K_4[Fe(CN)_6]$, $[Co(NH_3)_6]Cl_3$.

Complex compound:



Ligand: e^- pair donor atom or ion. Ex: $(CN)^-, Cl^-, Br^-, etc.$

→ Denticity: ability to donate e^- pairs.

① Mono dentate — forms 1 dative bond

Ex: -ve: ends with 'o' (or) 'ido'

F^- — fluoro

Cl^- — chlorido

Br^- — bromido

I^- — iodido

H^- — hydrido

N^{3-} — nitrido

N_3^- — azido

NH_2^- — amido

CH_3COO^- — acetato

O^{2-} — oxo

O_2^{2-} — peroxo

OH^- — hydroxo

NO_2^- — nitrito-N

ONO^- — nitrito-O

CN^- — cyano

NC^- — isocyano

SCN^- — thiocyano

NCS^- — isothiocyano

amidentate ligand

— 2 donor atoms. Contain both -ve & l.p.

+ve:

NO^+ — nitrosonium

NO_2^+ — Nitrosonium — sp

$NH_2NH_3^+$ — hydroazinium

Neutral -

H_2O - aqua

NH_3 - ammine

CO - carbonyl

C_5H_5N - pyridine

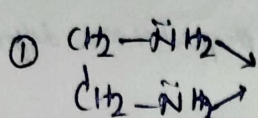
$H_2N-(CH_2-CH_2-NH_2)$ ethane-1,2-diamine

$NH_2(CH_3)$ - methylamine

I Bidentate ligands:

abbreviation

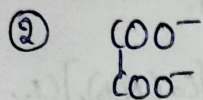
charge



(en)

ethylene diamine

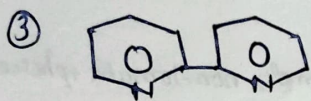
0



(ox)

oxalate ion

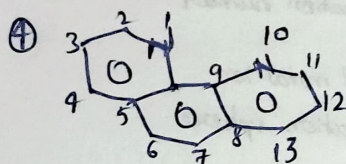
-2



(dipy)

1,2-dipyridine

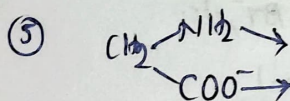
0



(phen)

1,10-phenanthroline

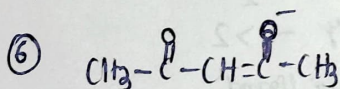
0



(gly)

glycerato ion

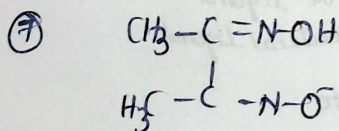
-1



(acac)

acetylacetonato

-1



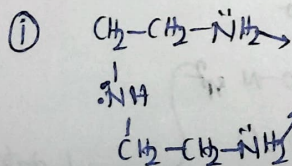
(dmg)

dimethyl glyoximate

-1

II Polydentate ligands

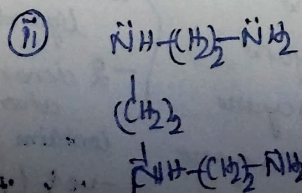
dentistry



(dien)

diethylene triamine

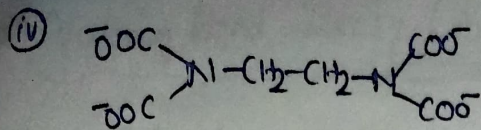
3



(tetraen)

1,3-bis(2-ethylamino)propane

4



EDTA

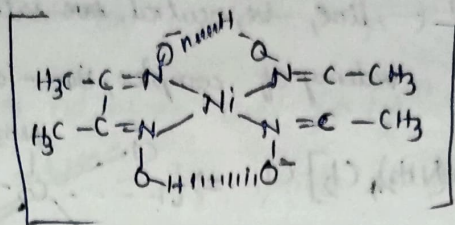
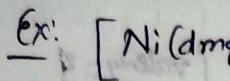
ethylenediamine
tetraacetate

6

→ Chelating ligands:

→ Ring formation by ligands mostly bidentate.

→ Rings ↑ stability ↑



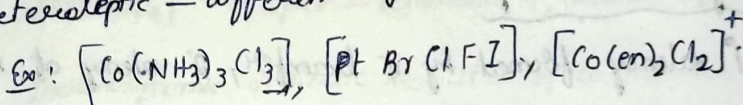
Cherry red complex.

→ Bidentate ligands:

Ligands show variable denticity.

Ex: SO_4^{2-} — 1 & 2, EDTA^{4-} show 4 & 6 denticity.

Complex $\left\{ \begin{array}{l} \text{Homoleptic} - \text{same ligands} \\ \text{Ex: } \text{K}_4[\text{Fe}(\text{CN})_6], [\text{Co}(\text{NH}_3)_6]\text{Cl}_3 \\ \text{Heteroleptic} - \text{different ligands:} \end{array} \right.$



→ Coordination polyhedron — spatial arrangement (3d space geometry)

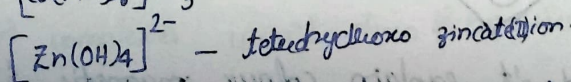
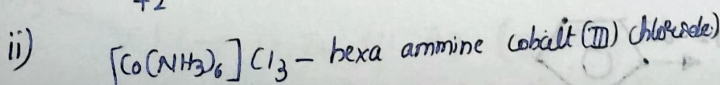
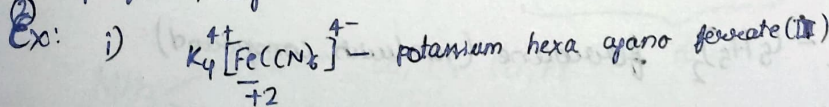
→ Mononuclear — 1 metal ion Ex: $\text{K}_4[\text{Fe}(\text{CN})_6]$, $[\text{PtCl}_4]^{2-}$

Poly nuclear — 2 or more metal ions Ex: $\text{Co}_2(\text{CO})_8$, $\text{Mn}_2(\text{CO})_{10}$

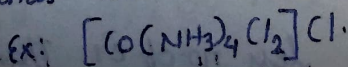
⇒ IUPAC Nomenclature

① i) +ve ion + prefix + ligand + C.M.A (ate) + O.S.C.

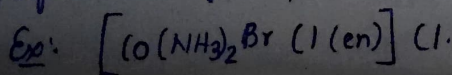
ii) prefix + ligand + C.M.A + O.S.C + counterion



② i) Ligands must be arranged acc to alphabetical order.



ii) for polydentate ligands, abbreviation should be considered.



⇒ Werner theory -

1st Valency - i) O.S of central metal ion

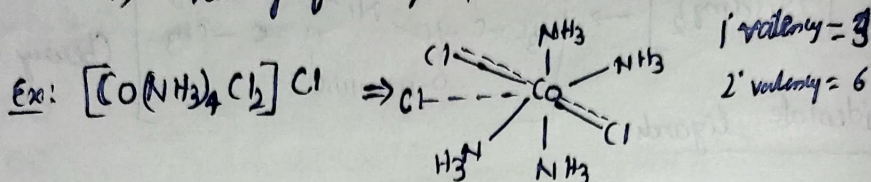
ii) $\rightarrow$ lines, only -ve ligands satisfy, ionic bond.

iii) don't involve in geometry, ionisable.

2nd valency - i) no of dative bonds

ii) $\leftarrow$ line, -ve, neutral, ~~are~~ satisfy through co-ordinate covalent bond.

iii) valency of complex, Non-ionisable.



Limitations - i) don't explain colour, magnetic & directional property of dative bonds.

ii) Why elements forms complexes?

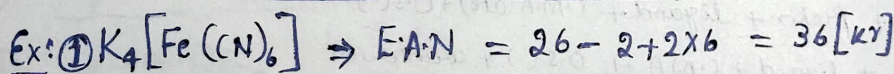
⇒ Sidgwick theory :

i) 1st valency formed by e^- transfer, 2nd valency of metal corresponds to no. of dative bonds.

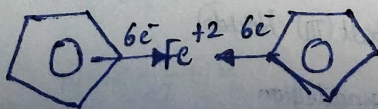
ii) central metal ion will continue accepting e^- from ligand to attain nearest noble gas config.

iii) total no. of e^- around $(M.A = E.A \cdot N)$.

$$E.A.N = Z - O.S + 2 \times C.No$$



② $\text{Fe}(\eta^5\text{-C}_5\text{H}_5)_2$ ferrocene (sandwich compound)



Limitations - i) don't explain colour, magnetic & stability of complexes.

ii) Why complexes with less no. of e^- than nearest noble gas are more stable?

⇒ Valency Bond Theory (Pauli)

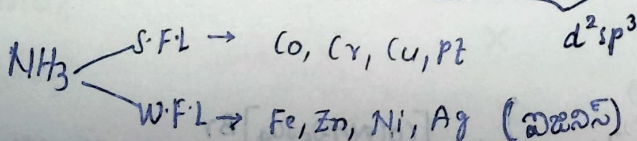
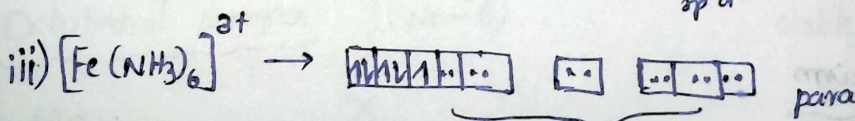
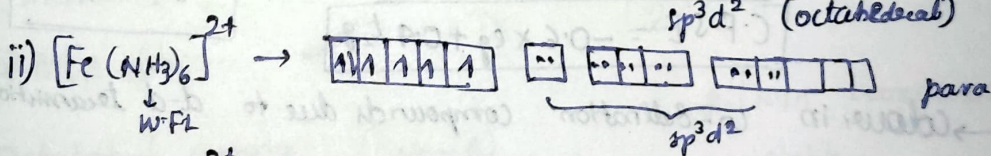
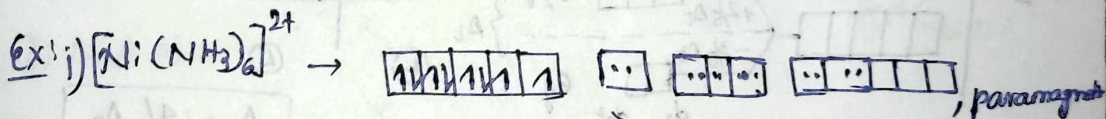
- C.M.A. does require e^- s & form ion by creating vacant orbitals (O.S.)
- In octahedral complexes $(n-1)d, ns, np$ used for hyb^n in presence of strong field ligand d^2sp^3 - inner orbital (&) low spin (&) spin paired (&) strong field (&) covalent complexes (pairing against Hund's rule).
- In octahedral complexes nd, ns, np used for hyb^n in presence of weak field ligand then sp^3d^2 - high spin (&) spin free (&) weak field (&) ionic complexes. (acc. to Hund's rule)
- Incompletely filled orbitals give magnetic moment.

for d^1, d^5, d^6, d^7 - ligand strength applied.

$d^1, d^2, d^3, d^8, d^9, d^{10}$ - ligand strength is not applied

Limitations - i) $[Cu(NH_3)_4]^{2+}$ expected - tetrahedral sp^3 , actual - sq. planar (d^9).

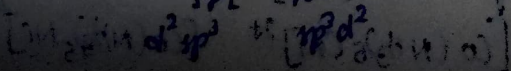
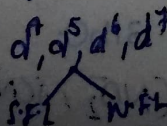
ii) thermodynamic & kinetic stabilities of coordination compounds, magnetic data & colour properties not explained



→ sp^3 & dsp^2 complexes won't show isomerism

Trick

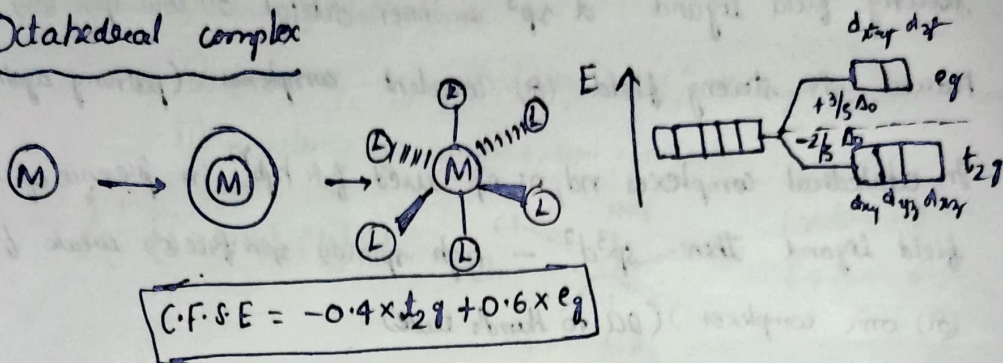
d^1, d^2, d^3 - S.F.L (&) W.F.L (always) d^2sp^3 hyb., d^8, d^9, d^{10} - S.F.L (&) W.F.L always sp^3d^2 hyb.



⇒ Crystal field theory

- Acc. to metal-ligand bond is ionic (due to electrostatic bonding)
- Ligands are point charges, anionic ligands are dipoles
- When ligand approaches C.M. ion d-orbital split into 2 sets

Octahedral complex



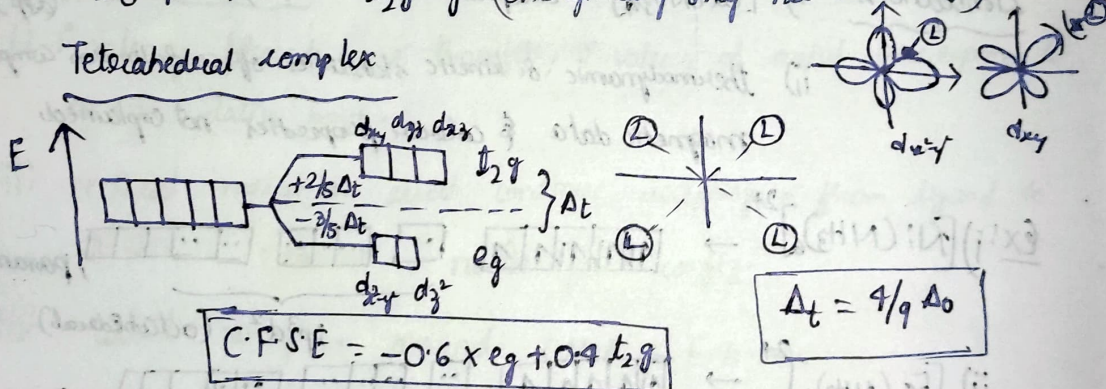
spectrochemical series — $I^- < Br^- < SCN^- < Cl^- < S^{2-} < OH^- < C_2O_4^{2-} < H_2O$

$NCS^- < edta^{4-} < Py, NH_3 < en < NO_2^- < CN^- < CO$

→ If $\Delta_o < P$ 4th $e^- \rightarrow t_{2g}^3 e_g^1$

$\Delta_o > P$ 4th $e^- \rightarrow t_{2g}^4 e_g^0$ (energetically only not kinetically)

Tetrahedral complex

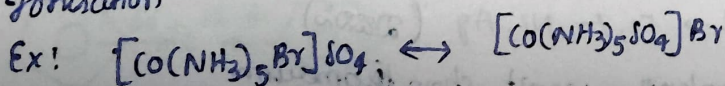


→ colour in co-ordination compounds due to d-d transition of e^-

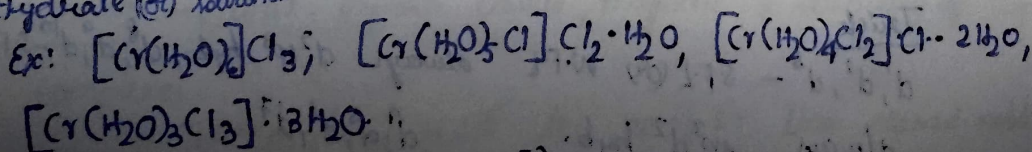
⇒ Isomerism

→ Structural

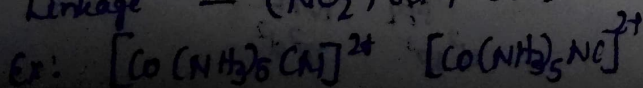
i) Ionisation



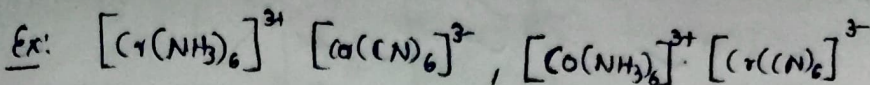
ii) Hydrate (solvation)



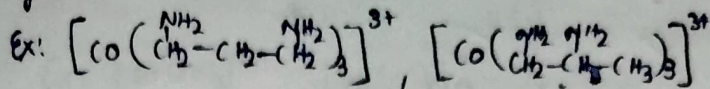
iii) Linkage — (NO_2^-, SCN^-, CN^-)



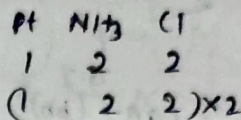
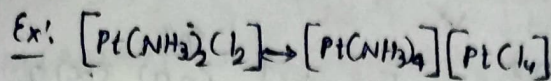
iv) Coordination isomerism



v) Ligand isomerism



vi) Polymerisation isomerism

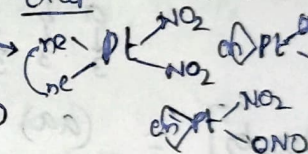


$\Rightarrow$ Stereoisomerism

square planar complex (C.No-4)

	$\frac{0 \cdot 1}{x}$	$\frac{0 \cdot 1}{x}$	$\frac{7 \cdot 1}{0}$
① Ma_4	x	x	0
② Ma_3b	x	x	0
③ Ma_2b_2	✓	x	2
④ Mab_2cd	✓	x	$3 = \frac{6}{2} + 1$
⑤ $\text{M}(\text{A}-\text{A})_2$	x	x	0
⑥ $\text{M}(\text{A}-\text{B})_2$	✓	x	2
⑦ $\text{M}(\text{A}-\text{A})\text{b}_2$	x	x	0
⑧ $\text{M}(\text{A}-\text{A})\text{bc}$	x	x	0
⑨ $\text{M}(\text{A}-\text{B})\text{b}_2$	x	x	0
⑩ $\text{M}(\text{A}-\text{B})\text{bc}$	✓	x	2

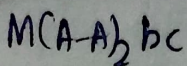
Except



Octahedral complex (C.No-6)

① Ma_6	x	x	0
② Ma_5b	x	x	0
③ Ma_4b_2	✓	x	2
④ Ma_3b_3	✓	x	2
⑤ $\text{Ma}_2\text{b}_2\text{c}_2$	✓	✓	$\frac{6}{5} + 1 = 6$
⑥ Mabcdef	✓	✓	$15 + 15 = 30$
⑦ $\text{M}(\text{A}-\text{A})_3$	✓	✓	2 $\rightarrow$ only 0.7
⑧ $\text{M}(\text{A}-\text{A})_2\text{b}_2$	✓	✓	$\frac{6}{2} + 1 = 3$

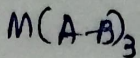
* trans-isomer is stable *



$$\frac{G.I}{\checkmark}$$

$$\frac{O.I}{\checkmark}$$

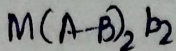
$$\frac{7.1}{3}$$



$$\checkmark$$

$$\checkmark$$

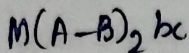
$$4$$



$$\checkmark$$

$$\checkmark$$

$$4$$

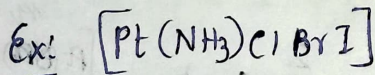
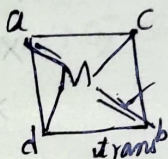
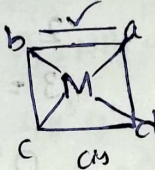
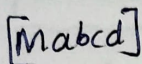
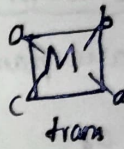
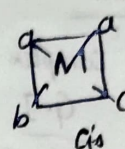
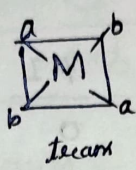
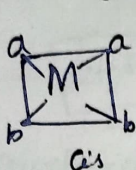
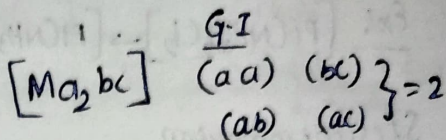
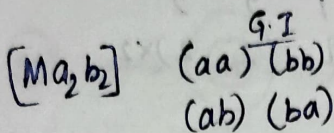


$$\checkmark$$

$$\checkmark$$

$$4$$

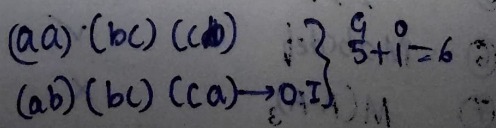
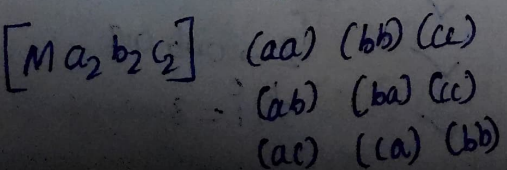
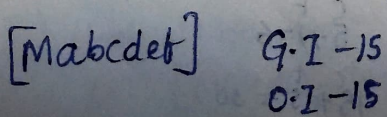
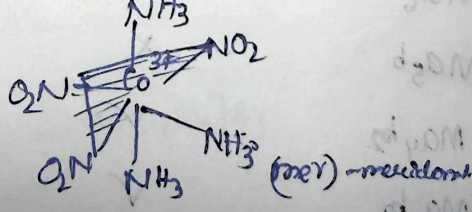
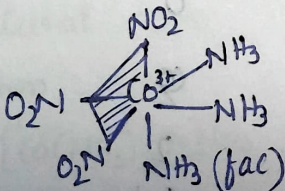
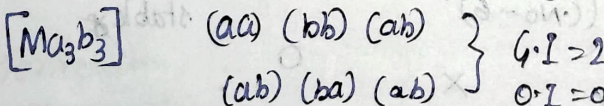
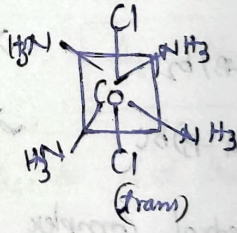
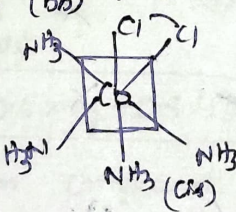
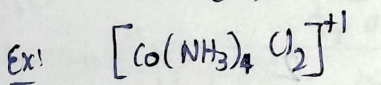
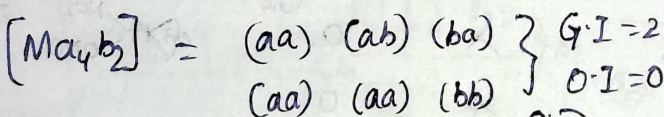
$$G.I < \begin{matrix} \text{dip}^2 \\ \text{sp}^3 \end{matrix} \quad \begin{matrix} \frac{S.I}{\checkmark} \\ \frac{O.I}{\times} \end{matrix}$$

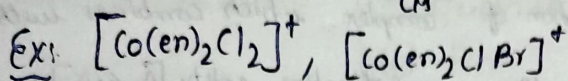
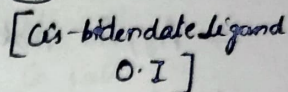
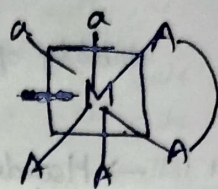
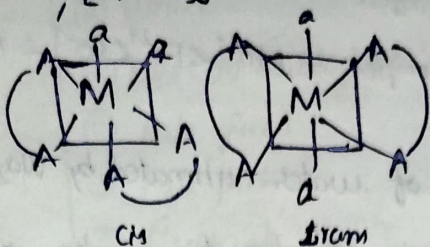
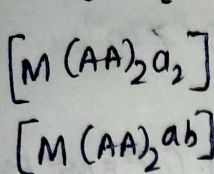
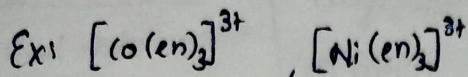
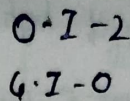
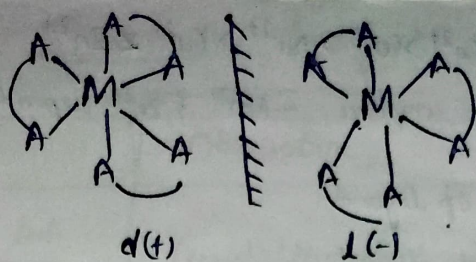
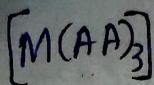


$$\text{No. of G.I} = 3$$

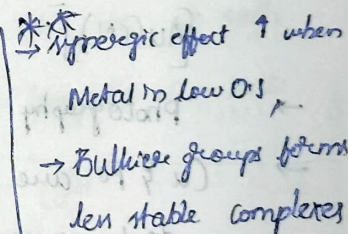
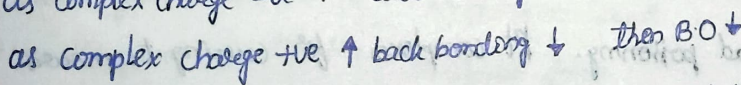
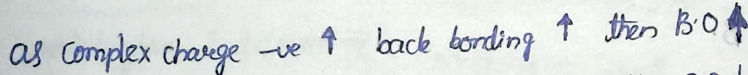
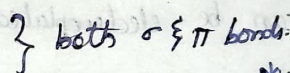
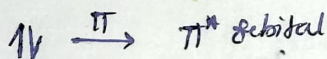
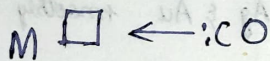
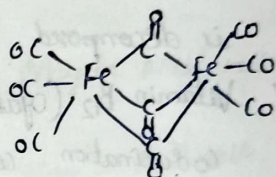
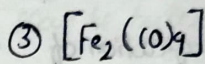
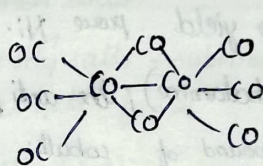
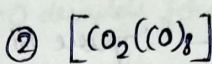
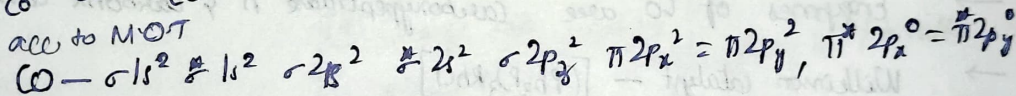
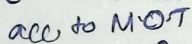
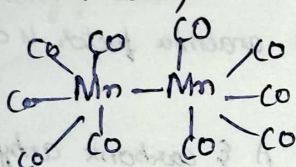
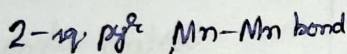
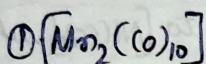
$$\text{total of I's} = 12$$

$$\begin{matrix} a & b & c & d \\ \frac{1}{3} & \frac{1}{3} & \frac{1}{3} & \frac{1}{3} \end{matrix}$$

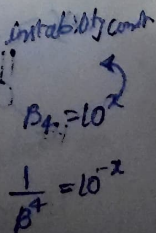
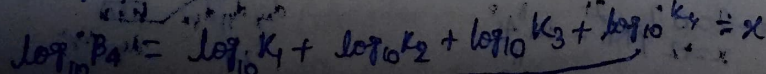
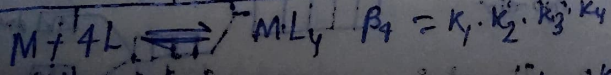
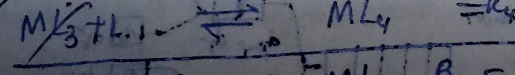
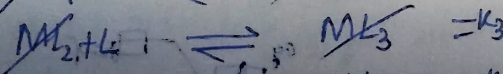
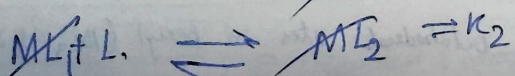




Metal carbonyls (Synergic effect)



Step wise stability const. & overall stability const.



→ Size order — $Mn^{2+} > Fe^{2+} > Co^{2+} > Ni^{2+} > Cu^{2+} > Zn^{2+}$

① stability of high spin complexes — Mn^{2+} & Zn^{2+} depends on HSE of ion

→ size & charge of ligand

→ basic nature of ligand

→ chelating effect

stability of complex — $Mn^{2+} < Fe^{2+} < Co^{2+} < Ni^{2+} < Cu^{2+} > Zn^{2+}$

Uses: → Hardness of water estimated by Na_2EDTA . (Ca^{2+} & Mg^{2+} form complex)

→ Ag, Au make use in formation of complex which combines with CN^- in presence of O_2 & H_2O to form co-ordination entity $[Au(CN)_2]$ in aq. soln.

Au can be separated in metallic form from this soln. by addⁿ of Zn.

→ Purification of metals — impure Ni is converted to $[Ni(CO)_4]$ which is decomposed to yield pure Ni.

→ Vitamin B_{12} (cyanocobalamin), the anti pernicious anaemia factor is a co-ordination compound of cobalt.

→ Enzymes of CO are carbonylpeptidase A & carbonic anhydrase.

→ Wilkinson catalyst — $[C(PH_3)_3RhCl]$

→ Articles can be electroplated with Ag & Au smoothly $[Ag(CN)_2]^-$ & $[Au(CN)_2]^-$.

→ Photography $AgBr \rightarrow [Ag(S_2O_3)_2]^{3-}$

→ Cu & Fe are removed from body by D-penicillamine & desferrioxamine B.

→ EDTA for lead poisoning. Pt complexes for curing tumours.

Cu $[Pt(NH_3)_2Cl_2]$ for cancer.

→ Ruby is Al_2O_3 about 0.5-1%. Cr^{3+} ions and randomly occupied by Al^{3+} . Emerald Cr^{3+} occupies octahedral sites in beryl ($Be_3Al_2Si_6O_{18}$)

→ $[Ni(dmg)_2]$

